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Original Article

Dual-Frequency Bioelectrical Impedance Analysis is Accurate and Reliable to Determine Lean Muscle Mass in The Elderly

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Abstract

Background: Dual-frequency bioelectrical impedance analysis (DF-BIA) devices are more accessible and affordable than dual-energy X-ray absorptiometry (DXA); however, no studies have reported the accuracy of DF-BIA in body composition measurement, especially in the Thai elderly. The aims of this study were to (1) compare the accuracies of lean muscle masses measured by DF-BIA devices and DXA and (2) assess the reliability of the DF-BIA device. *Methods:* This cross-sectional study was conducted on participants older than 60 years who visited the Orthopedic Clinic of Siriraj Hospital. Whole-body and appendicular skeletal muscle masses (ASMs) were measured using DF-BIA (Tanita RD-545), with DXA (GE Lunar iDXA) as the standard reference. The test-retest reliability of the DF-BIA and the agreement between the devices were assessed using the intraclass correlation coefficient (ICC) and Bland–Altman plots. Regression analysis was used to develop an equation to estimate ASM values from BIA close to those from DXA. *Results :* The mean age of 88 participants was 73.8 (SD 8.0) years, with women predominating (84.1%). The agreement of BIA and DXA was very high for whole-body lean mass (ICC = 0.954) and ASM (ICC = 0.954), but the mean difference in muscle mass from DF-BIA was overestimated. The ICCs of test-retest reliability for whole-body muscle mass and ASM were 0.987 and 0.988, respectively. The equation for corrected ASM was formulated from a linear equation ($R^2 = 0.93$). *Conclusions:* Although lean muscle mass from DF-BIA was minimally overestimated relative to DXA, this device had high accuracy and reliability for lean muscle mass evaluation in the elderly. DXA and DF-BIA are interchangeable for the assessment of muscle mass.

Key Words: Dual-frequency; Bioelectrical impedance analysis; Appendicular skeletal muscle mass; Dual-energy X-ray absorptiometry; Elderly.

Introduction

Sarcopenia is a skeletal muscle disorder characterized by a generalized and progressive decrease in skeletal muscle mass and muscle strength, and it is common in the elderly population *(1)*. The disease is associated with an increased risk of undesired outcomes, such as falls, fractures, disability, and mortality *(2–5)*. Regarding financial aspects, sarcopenia is costly for healthcare systems, leading to financial burdens when left untreated *(6)*. Using affordable, accessible, and reliable diagnostic devices can improve physician awareness of sarcopenia and promote optimal care.

According to the European Working Group on Sarcopenia in Older People and Asian Working Group on Sarcopenia, the parameters for diagnosing sarcopenia are

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muscle strength, skeletal muscle mass, and physical performance *(7,8)*. Skeletal muscle mass or quantity can be measured using magnetic resonance imaging, dual-energy X-ray absorptiometry (DXA), and bioelectrical impedance analysis (BIA). DXA is most widely used for standard reference measurements of lean muscle mass *(9)*. The advantages of DXA are that it is noninvasive, delivers highly reliable results, and provides reproducible estimates of skeletal muscle mass. However, the main disadvantage of DXA is that it lacks portability (due to the size of the device), making it inconvenient to use in daycare orthopedic clinics. Moreover, the device is costly and needs ongoing maintenance, restricting its availability at hospitals. The use of DXA also involves radiation exposure and requires highly trained personnel.

Another device commonly used is BIA, which can measure lean muscle mass indirectly using the electrical conductivity principle. BIA devices are inexpensive, portable, and easy to perform *(9)*. Due to these advantages, these devices may be interchangeable with DXA *(8)*. On the other hand, different brands of BIA may give inconsistent results because of differences in the equations used to estimate muscle mass *(10)*. The results of BIA are also related to age and race or ethnicity *(11)*. Consequently, it is essential to compare the accuracy of specific brands of BIA with the corresponding DXA values for each population.

The accuracy of multifrequency bioelectrical impedance analysis (MF-BIA) in measuring lean muscle mass has been demonstrated in several populations *(12,13)*. Dual-frequency bioelectrical impedance analysis (DF-BIA) devices are more accessible, affordable, easier to perform, and able to provide comprehensive measurements of body composition in less than a minute. Due to these advantages, dual-frequency BIA can be used in the outpatient clinics of resource-limited local hospitals, during home visits, or as part of national surveys. Although there was only one study that explored the accuracy of DF-BIA for body composition measurement compared to DXA *(14)*, no study reports the accuracy and reliability of DF-BIA in the elderly Asian population. The current investigation focused on 2 objectives. First, it compared the accuracies of lean muscle mass measurements of Thai elderly individuals obtained with a DF-BIA device and DXA (as the standard reference). Second, it evaluated the reliability of the DF-BIA device.

Materials and methods

Before this cross-sectional study began, its protocol was approved by the Siriraj Institutional Review Board of the Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand (COA number Si 692/2021). The study was carried out on participants aged 60 years or older who visited the Orthopedic Outpatient Department of Siriraj Hospital, Mahidol University, Bangkok, between September 2021 and February 2022. We excluded participants if they had an electronic implant (e.g., a heart pacemaker or brain stimulator), had a body mass index (BMI) greater than 34 kg/m^2 (which might lead to results being over- or underestimated) *(15)*, or had undergone limb amputation. DF-BIA and DXA were used to measure each participant's whole-body and segmental lean-muscle masses on the same day during the early morning hours. The enrolled participants were instructed to fast for at least 3 hours and avoid drinking for at least 1 hour before the measurements *(16)*. Informed consent was obtained prior to inclusion in the study.

Body composition measurements

Dual-energy X-ray absorptiometry

The DXA measurements were performed with a total body scan (General Electric Healthcare, Lunar iDXA). It incorporated a constant potential X-ray source of 100 kV and was calibrated with a phantom daily. DXA was used as the reference method to measure whole and regional body composition, including fat, bone mineral density, and lean muscle mass. The participants were positioned according to the manufacturer's instructions (supine position on the scanner table, legs straight, arms on their sides, and stationary during the scan). The participants were also required to wear light clothing to minimize cloth absorption. Well-trained and specialized radiology technologists performed the measurements. The lean muscle mass was assessed with DXA analysis software (Encore software version 16 SP1) to obtain data for different regions of interest (the trunk, arms, and legs) that would be used for later comparisons with the DF-BIA analysis.

Bioelectrical impedance analysis

A dual-frequency BIA (RD-545; Tanita Corporation, Tokyo, Japan) was used for the body composition and lean muscle mass measurements. This device incorporates dual-frequency technology and provides acceptable body composition estimates. It is safe as it passes a low-level electrical current. Whole-body and appendicular skeletal muscle masses (ASMs; left arm, right arm, left leg, and right leg) were each measured using 2 frequencies (50 kHz and 6.25 kHz). The muscle mass measurements were obtained following the manufacturer's instructions. The participants were required to remove their shoes and socks or stockings. The soles of their feet and the palms of their hands were then cleaned. They were instructed to stand upright on footpads, extend their arms away from their chests, and hold the handles of the BIA device. Measurements were taken twice at 5-minute intervals. The results of each participant were displayed on a monitor and recorded. A well-trained research assistant carried out the BIA measurements.

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Anthropometric measurements

Before the DXA and DF-BIA measurements, standing height was measured to the nearest 0.01 m using a stadiometer, and body weight was measured to the nearest 0.01 kg using a weighing scale. BMI was calculated as body weight in kilograms divided by height in meters squared. Calf circumference was measured with a standard measuring tape at the point of the largest circumference of the dominant side.

Statistical analysis

A Shapiro–Wilk normality test was used to determine the normal distribution for all parameters. Continuous variables are reported as the means ± SDs or median with interquartile range for nonsymmetrical distributions. Categorical variables are reported as numbers and percentages. Differences in lean muscle mass measurements obtained with DF-BIA and DXA were compared using Student's t-test or Wilcoxon signed-rank test, as appropriate. The level of agreement between the 2 devices was assessed using the intraclass correlation coefficient (ICC) and the Bland–Altman method *(17)*. A correlation coefficient close to 1 represented high reliability between the 2 devices. We considered an ICC ≥ 0.90 as very high, between 0.70 and 0.89 as high, and between 0.50 and 0.69 as moderate *(18)*. Systemic bias was evaluated as the mean difference between the 2 methods through Bland–Altman plots. The limits of agreement were calculated as the bias ± 2SD of the difference between the methods. Moreover, the test-retest reliability of the DF-BIA device was determined using ICC. For clinical diagnoses of sarcopenia, a linear regression analysis was used to formulate a population-specific equation that can estimate ASM and ASM divided by height in meters squared (ASM/height^2) values close to those obtained with DXA. Statistical analyses were performed using PASW Statistics for Windows, version 18.0 (SPSS Inc., Chicago, IL, USA) or MedCalc for Windows, version 17.6 (MedCalc Software, Ostend, Belgium). Statistical significance for all tests was set at a probability (P) value of < 0.05.

The sample size of this study was calculated using statistical software packages (PASS version 14. NCSS, LLC. Kaysville, Utah, USA, ncss.com/software/pass) including the sample size estimators for the Bland-Altman method. For the expected mean of differences and standard deviation of differences input, we used the mean and standard deviation difference of appendicular lean mass measured by DXA and BIA devices from the previous study *(18)*, accounting for 1.75 and 0.5, respectively. The maximum allowed difference between the 2 methods was 3 kg. Given a 95% confidence interval and probability of making a type II error of 20%, a sample of at least 86 participants was determined to be required.

Results

The demographic and clinical characteristics of the participants are summarized in Table 1. The average age of the 88 participants was 73.78 years, and women predominated (84.1%). The average BMI and leg circumference were 24.1 kg/m^2 and 33.01 cm, respectively.

The body compositions measured by DXA and DF-BIA are detailed in Table 2. For lean muscle mass measured by DXA, the mean whole-body mass was 33.11 ± 6.91, and the ASM was 14.52 ± 3.54 kg. For lean muscle mass measured by DF-BIA, the mean whole-body mass was 34.18 ± 7.01 kg, and the ASM was 15.02 ± 3.53 kg. When comparing DF-BIA with DXA, the lean muscle mass values obtained from DF-BIA were higher; however, there was no significant difference between the 2 methods for whole-body ($P = 0.292$) or ASM ($P = 0.180$). Lean muscle mass values of upper extremities obtained from DF-BIA were significantly lower, whereas values of lower extremities from DF-BIA were greater. The ASM/height^2 was also calculated and analyzed to determine its

Table 1
Demographics and clinical characteristics.

Clinical variable	Mean ± SD or Number (%) (N = 88)
Age (years)	73.8 ± 8.0
Sex; female	74 (84.1)
Height (meters; m)	1.52 ± 0.09
Weight (kilograms; kg)	56.12 ± 13.18
Body mass index (kg/m^2)	24.06 ± 4.60
Calf circumference (centimeters; cm)	33.08 ± 3.68

Table 2
Comparison of average body composition measured by DF-BIA and DXA.

Measurement	DXA	DF-BIA	P value
Whole-body muscle mass (kg)	33.11 ± 6.91	34.18 ± 7.01	0.790
ASM (kg)	14.52 ± 3.54	15.02 ± 3.53	0.180
Left arm	1.72 ± 0.48	1.52 ± 0.36	0.001
Right arm	1.80 ± 0.50	1.56 ± 0.35	< 0.001
Left leg	5.45 ± 1.33	5.96 ± 1.44	0.003
Right leg	5.55 ± 1.40	5.99 ± 1.48	0.013
ASM/height^2 (kg/m^2)	6.20 ± 1.07	6.42 ± 1.07	0.135
Percentage body fat (%)	35.13 ± 6.92	34.71 ± 8.26	0.664

ASM, appendicular skeletal muscle mass; DF-BIA, dual-frequency bioelectrical impedance analysis; DXA, dual-energy X-ray absorptiometry.

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Table 3
Agreement between DF-BIA and DXA for body composition measurements.

Measurement	ICC	95% CI
Whole-body muscle mass (kg)	0.954	0.97–0.99
ASM (kg)	0.954	0.90–0.98
Left arm	0.895	0.85–0.93
Right arm	0.868	0.81–0.91
Left leg	0.936	0.91–0.96
Right leg	0.926	0.89–0.95
ASM/height^2 (kg/m^2)	0.908	0.81–0.95
Percentage body fat (%)	0.876	0.82–0.92

ASM, appendicular skeletal muscle mass; CI, confidence interval; DF-BIA, dual-frequency bioelectrical impedance analysis; DXA, dual-energy X-ray absorptiometry; ICC, intraclass correlation coefficient.

clinical utility for diagnosing sarcopenia. The ASM/height^2 measured by DF-BIA was 6.42 ± 1.07 kg/m^2, while that measured by DXA was 6.20 ± 1.07 kg/m^2. No significant differences between the 2 methods were observed ($P = 0.135$). Between the BIA and DXA measurements, the agreement of whole-body muscle mass was very high. Similarly, the ASMs of the arms and legs showed a high to a very high agreement. ICC values with 95% confidence intervals are presented in Table 3.

The differences in whole-body muscle mass and ASM obtained by the 2 methods were assessed using Bland–Altman plots (Fig. 1). Most individual plots were within the 2SD range, ranging from -4.9 to 3.2 for whole-body muscle mass and from -2.4 to 1.4 for ASM. On average, the whole-body muscle mass and ASM values from BIA appeared to be overestimated by 0.9 kg and 0.5 kg, respectively, relative to DXA, without systemic bias.

Table 4
Test-retest reliability of the DF-BIA device.

Measurement	ICC	95% CI
Whole-body muscle mass (kg)	0.987	0.979–0.992
ASM (kg)	0.988	0.981–0.992
Left arm	0.970	0.953–0.981
Right arm	0.960	0.937–0.974
Left leg	0.923	0.882–0.951
Right leg	0.968	0.950–0.980
ASM/height^2 (kg/m^2)	0.981	0.971–0.988
Percentage body fat (%)	0.998	0.997–0.999

ASM, appendicular skeletal muscle mass; CI, confidence interval; DF-BIA, dual-frequency bioelectrical impedance analysis; ICC, intraclass correlation coefficient.

The ICCs of test-retest reliability for whole-body muscle mass and ASM were 0.987 and 0.988, respectively. The ASMs also demonstrated a high degree of reproducibility, with ICCs of more than 0.90 (Table 4).

Simple linear regression analysis was used to formulate equations to correct for the overestimates of ASM and ASM/height^2 measured by BIA so that they were close to the corresponding DXA values. The 2 equations are:

$$ASM_{DXA} = 0.02 + 0.97 \times \mathrm{ASM_{BIA}} (\mathrm{P} < 0.001;\ R^2 = 0.93)$$

$$ASM/height^2_{DXA} = 0.26 + 0.92 \times \mathrm{ASM/height^2_{BIA}} (\mathrm{P} < 0.001;\ R^2 = 0.86).$$

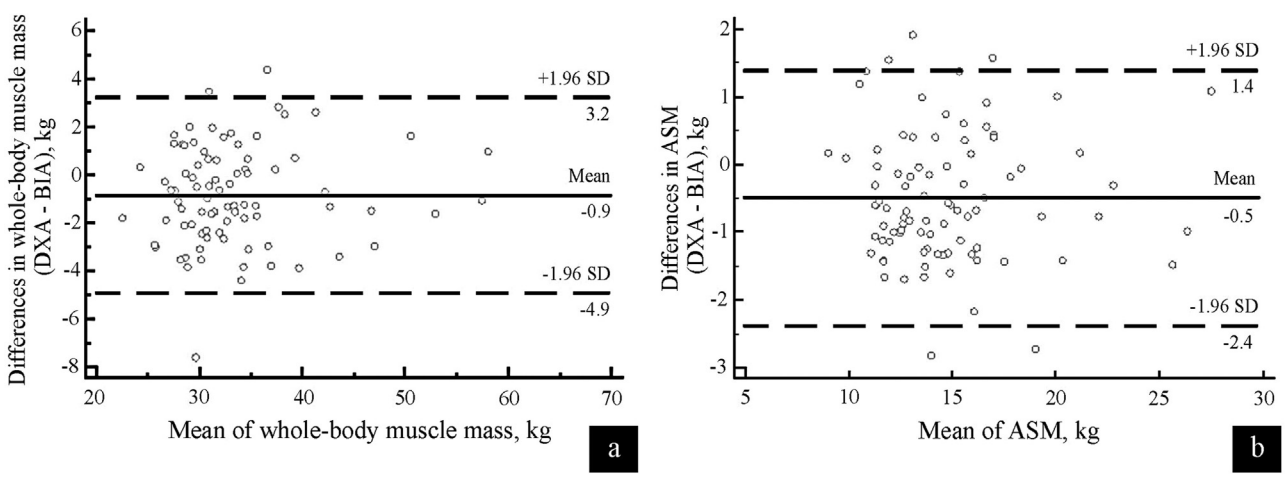

Fig. 1. Bland–Altman plots with mean differences and 95% limits of agreement for lean muscle mass measured by DXA and BIA. (a) whole-body muscle mass (b) appendicular skeletal muscle mass.

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Discussion

This study demonstrated the accuracy of DF-BIA in the measurement of muscle mass in elderly individuals using DXA as a reference standard. The portable DF-BIA device is interchangeable with DXA for muscle mass assessments. For whole-body muscle mass and ASM, the results showed a high degree of agreement between the 2 methods and high test-retest reliability of DF-BIA for both whole-body and segmental muscle mass. However, the ASM measured by BIA tended to be slightly overestimated relative to the values obtained with DXA.

The results of the present study showed excellent agreement between the 2 methods for muscle mass measurement; however, it was inconsistent with the a previous study of Gade et al which reported that DF-BIA and DXA could not be used interchangeably in the elderly *(14)*. The possible explanations for the divergent outcomes were the differences in study settings, participants' characteristics, and DF-BIA manufacturers *(19)*. These results are compatible with several studies investigated the MF-BIA accuracy and reliability. Previous studies on the Asian population revealed high accuracy of MF-BIA for body composition measurement compared to DXA *(12,19,20)*; however, research on Caucasian revealed a low accuracy of MF-BIA *(18,21,22)*. These disparate results support the proposition that a population-specific database of muscle mass measured by BIA should be obtained. Additionally, an equation for converting DF-BIA values to DXA values should be developed for the Thai population. Our linear regression analysis found no significant independent variables (age, sex, height, weight, BMI, and calf circumference). Consequently, the 2 formulated equations presented in this paper are simple linear equations with a high coefficient of determination.

Additionally, the ASM and ASM/height^2 obtained by DF-BIA were overestimated compared to those obtained by DXA. This finding was consistent with previous studies: BIA overestimated the mean ASM/height^2 by 1.5 kg in the Korean population *(12)* and 1.75 kg in the Belgian population *(18)*. However, the difference in mean ASM/height^2 was much lower in our results (0.22 kg). This lower value could be explained by differences between the studies regarding their BIA analyses and the software they used. In the current work, BIA underestimated the percentage of body fat compared to the values obtained by DXA, which may have been due to the female sex predominating. A similar result was found in the study by Gupta et al, who reported an underestimation of percentage body fat in females *(23)*.

The mean ASM measured by DXA in this study was 14.52 ± 3.54 kg. This value was less than the cutoff points for a low muscle quantity in men (20 kg) and women (15 kg) *(8)*. The results imply that Thai elderly individuals with orthopedic problems seem to have low muscle mass; thus, there is a high probability of diagnosing sarcopenia. We suggest routine screening for sarcopenia in elderly individuals suffering from orthopedic problems. Nevertheless, using DXA for muscle mass measurements in all patients can result in high hospital costs and disrupt the provision of an essential service, especially for bone mineral density measurements. Moreover, DXA is not portable and is only found in hospitals. Therefore, it is inconvenient to use the device on older adults living in remote areas.

The results of our study showed that the DF-BIA device has high accuracy and reliability for muscle mass evaluations and can be interchangeable with DXA to avoid the problems mentioned above. Although many previous studies have compared body composition measurements using DXA and MF-BIA, this is the first study to use DF-BIA in elderly Asian individuals. DF-BIA has many advantages over MF-BIA, such as being less expensive, portable (due to its lower weight and smaller size), and requiring little training for test administrators. DF-BIA may be an optimal method for detecting sarcopenia, especially in nursing homes, small health centers, and household surveys.

Aside from sarcopenia screening, using DF-BIA for muscle mass evaluations may be applied for other purposes, including exercise program modification, diet planning, or other lifestyle adjustments. Further investigation is warranted to determine whether muscle mass measured by the DF-BIA device can be utilized in lifestyle modification protocols as a monitoring tool. Moreover, more studies are encouraged to ascertain whether using DF-BIA can improve long-term compliance. Since DF-BIA can be performed considerably more quickly than DXA, DF-BIA is likely to be used much more frequently and hence may encourage compliance better.

This study has some limitations. First, since the data were obtained from elderly participants, the equation for ASM conversions cannot be applied to the young adult population. However, low muscle mass in adults occurs only in susceptible patients, especially those with obesity and metabolic syndrome, who prefer DXA for evaluations *(24)*. In addition, the included participants were predominantly elderly women. Women are more likely to have lower lean muscle mass than men *(25)*. The predominance of women in the trial may prevent generalization to men. Second, we did not evaluate sarcopenia in this study, although using the DF-BIA device to assess lean muscle mass in our study revealed that one-fourth of the subjects were at risk for sarcopenia. Further study should be conducted to evaluate the reliability and validity of the DF-BIA device specifically for the diagnosis of sarcopenia. Third, the participants' hydration status and physical activity were not controlled before the BIA measurement and may have influenced the results. Moreover, the participants' nutritional status was not assessed. However, all patients were outpatients without critical illness or severe comorbidities, and their mean BMI was 24.06 kg/m^2. Thus, we assumed that there was no malnutrition in the study cohort. Fourth, this study assessed the reliability of

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DF-BIA for muscle mass from 2 measurements on the same day. However, if the patient's hydration status is altered during the day, the results may differ, which might affect the reliability of the measurements. Last, this study used DXA as a reference measurement; however, different densitometer devices and software versions can report different body compositions *(26)*.

This study is the first to compare DXA and DF-BIA measurements of elderly individuals. For measuring muscle mass, the TANITA RD-545 DF-BIA device was in excellent agreement with standard DXA, although lean muscle mass from BIA was minimally overestimated relative to DXA. This device has high accuracy and reliability for lean muscle mass evaluations in elderly Thai people and is interchangeable with DXA.

Disclosure statement

The authors declare no conflicts of interest.

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